

EOD Document – Jenkins Pipeline Skeleton

Date: 06 October 2026

Task: Write Sample Skeleton of Pipelines

Technology: Jenkins Pipeline

Pipeline Types: Declarative and Scripted

1. Objective

The objective of this task was to understand and write the **basic skeleton/structure of Jenkins Pipelines**.

Jenkins provides two main Pipeline syntax types:

1. **Declarative Pipeline**
2. **Scripted Pipeline**

2. Declarative Pipeline Skeleton

The basic structure of a Declarative Pipeline is:

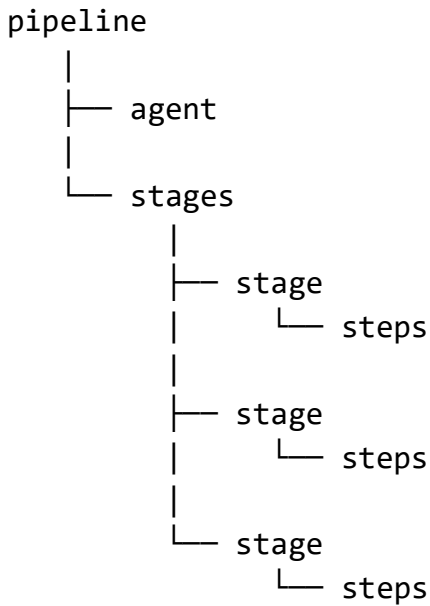
```
pipeline {  
  
    agent any  
  
    stages {  
  
        stage('Build') {  
            steps {  
                echo 'Build'  
            }  
        }  
  
        stage('Test') {  
            steps {  
                echo 'Test'  
            }  
        }  
  
        stage('Deploy') {
```

```

        steps {
            echo 'Deploy'
        }
    }
}

```

Structure



Important keywords

Keyword	Purpose
pipeline	Starting point of Declarative Pipeline
agent	Defines where the pipeline runs
stages	Container for all stages
stage	Defines an individual stage
steps	Contains commands executed in the stage

3. Scripted Pipeline Skeleton

The basic structure of a Scripted Pipeline is:

```

node {

    stage('Build') {
        echo 'Build'
    }
}

```

```

    stage('Test') {
        echo 'Test'
    }

    stage('Deploy') {
        echo 'Deploy'
    }
}

```

Structure

```

node
|
├── stage('Build')
|
├── stage('Test')
|
└── stage('Deploy')

```

Important keywords

Keyword	Purpose
node	Allocates a Jenkins executor/agent
stage	Defines a logical stage
echo	Prints a message in the Jenkins console
sh	Executes shell commands

4. Difference Between Declarative and Scripted

Declarative	Scripted
Starts with pipeline {}	Usually starts with node {}
Structured syntax	Groovy-based syntax
Easier to read	More flexible
Uses stages and steps blocks	Stages can be directly inside node
Good for standard CI/CD	Useful for complex programming logic

Declarative

```
pipeline {
  agent any

  stages {
    stage('Build') {
      steps {
        echo 'Build'
      }
    }
  }
}
```

Scripted

```
node {
  stage('Build') {
    echo 'Build'
  }
}
```

5. Sample CI/CD Structure

A real CI/CD pipeline can contain stages such as:

Git Checkout

↓

Build

↓

Test

↓

SonarQube

↓

Quality Gate

↓

Nexus

↓

Deploy

↓

Notification

This is only the **skeleton/structure**. Actual commands are added inside each stage depending on the project.

6. Lab Performed

The following concepts were practiced:

- Jenkins Pipeline structure
- Declarative Pipeline skeleton
- Scripted Pipeline skeleton
- node
- pipeline
- agent
- stages
- stage
- steps
- Basic echo commands
- Difference between Declarative and Scripted syntax

7. Result

Successfully understood the basic skeleton of Jenkins Pipelines and the structural differences between **Declarative** and **Scripted Pipelines**.

The main structures to remember are:

Declarative:

```
pipeline
→ agent
→ stages
  → stage
    → steps
```

Scripted:

```
node
→ stage
  → commands
```

